

LAPORAN PRAKTIKUM 8
SISTEM BASIS DATA



Disusun Oleh :

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Kelas Praktikum/Semester :

C/III

PROGRAM STUDI SISTEM INFORMASI
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A. Dasar Teori

MongoDB adalah sistem basis data NoSQL yang menggunakan model penyimpanan berbasis dokumen dengan format BSON sehingga mampu menangani data terstruktur maupun tidak terstruktur secara fleksibel. Melalui layanan cloud MongoDB Atlas, pengguna dapat membuat dan mengelola cluster tanpa instalasi manual, termasuk pengaturan whitelist IP dan pembuatan database user untuk menjaga keamanan akses. Atlas menyediakan kemudahan dalam proses deployment, monitoring, serta konektivitas melalui connection string `mongodb+srv://` yang digunakan sebagai alamat utama untuk menghubungkan aplikasi ke database.

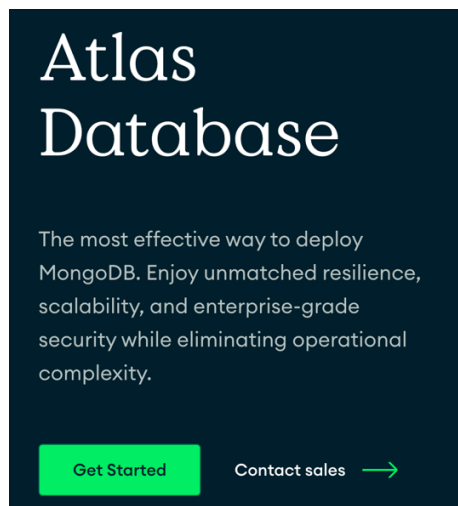
Dalam pengembangan aplikasi menggunakan Node.js, Express.js berfungsi sebagai framework untuk membangun server dan endpoint API, sedangkan Mongoose digunakan sebagai ODM untuk mengatur model data serta berinteraksi dengan MongoDB secara lebih terstruktur. Integrasi antara Express.js dan MongoDB Atlas memungkinkan pelaksanaan operasi database (CRUD) secara efisien. Untuk memastikan koneksi dan fungsi API berjalan dengan baik, Postman digunakan sebagai alat pengujian melalui berbagai permintaan HTTP seperti GET, POST, PUT, dan DELETE.

B. Langkah – Langkah Praktikum

Step 1: Membuat Akun MongoDB Atlas

1. Mendaftar Akun

Klik tombol “Try Free” atau “Get Started Free”, kemudian login menggunakan akun google



2. Menyelesaikan Registrasi



Welcome to Atlas. Let's build something great.

Help us tailor your experience by taking a minute to answer the questions below.

GETTING TO KNOW YOU

What is your primary goal?

Build a project I have in mind

How long have you been developing software with MongoDB?

I've never developed software with MongoDB before

GETTING TO KNOW YOUR PROJECT

What programming language are you primarily building on MongoDB with?

JavaScript / Node.js

What type(s) of data will your project use?

You can choose as many as you want

Customer... x User acti... x Time seri... x Catalog... x Streaming... x Location... x

Will your application include any of the following architectural models?

You can choose as many as you want

IoT / edge... x Mobile x AI/ML Enr... x Search en... x Distribut... x Serverless... x

[Skip personalization](#)
[Finish](#)

Step 2: Membuat Cluster

1. Membuat Proyek Baru

Pada dashboard, klik **“New Project”**, masukkan nama proyek, lalu klik **Next**.

?

6M

New Project

Create a Project

Name Your ProjectAdd Members

Name Your Project

Project names have to be unique within the organization (and other restrictions).

MuhammadHaritsArozzaq-240006087

Add Tags (Optional)

Use tags to efficiently label and categorize your projects. A project can have a maximum of 50 tags. You can modify tags for the project later. [Learn more](#)

Key	Value	Action
Select a key or enter your own	:	Select a value or enter your own
+ Add Tag		

CancelNext

2. Konfigurasi Cluster

Deploy your cluster

Use a template below or set up advanced configuration options. You can also edit these configuration options once the cluster is created.

M10

\$0.11/hour

Dedicated cluster for development environments and low-traffic applications.

STORAGE

10 GB

RAM

2 GB

vCPU

2 vCPUs

Flex

From \$0.011/hour

Up to \$30/month

For development and testing, with on-demand burst capacity for unpredictable traffic.

STORAGE

5 GB

RAM

Shared

vCPU

Shared

Free

For learning and exploring MongoDB in a cloud environment.

STORAGE

512 MB

RAM

Shared

vCPU

Shared

Free forever! Your free cluster is ideal for experimenting in a limited sandbox. You can upgrade to a production cluster anytime.

Configurations

Name

You cannot change the name once the cluster is created.

harits

Provider

aws

Google Cloud

Azure

Region

Jakarta (asia-southeast2)

★ Recommended ⓘ Low carbon emissions ⓘ

Tag (optional)

Create your first tag to categorize and label your resources; more tags can be added later. [Learn more.](#)

Select or enter key

Select or enter value

Quick setup

- ☒ Automate security setup ⓘ
- ☒ Preload sample dataset ⓘ

Step 3: Mengatur Whitelist IP Address

IP Address	Comment	Status	Actions
0.0.0.0/0 (includes your current IP address)		Active	EDIT DELETE

Step 4: Membuat Database User

Add New Database User

Create a database user to grant an application or user access to databases and collections in your clusters in this Atlas project. Granular access control can be configured with default privileges or custom roles. You can grant access to an Atlas project or organization using the corresponding [Access Manager](#).

Authentication Method

Password

Certificate

AWS IAM

Federated Auth (MongoDB 7.0 and up) ⓘ

MongoDB uses SCRAM as its default authentication method.

Password Authentication

harits-2400016087

SHOW

Autogenerate Secure Password

Copy

User Description

Add an optional description to your user.

Database User Privileges

Configure role based access control by assigning database user a mix of one built-in role, multiple custom roles, and multiple specific privileges. A user will gain access to all actions within the roles assigned to them, not just the actions those roles share in common. You must choose at least one role or privilege. [Learn more about roles.](#)

Built-in Role

Select one built-in role for this user.

Read and write to any database

Custom Roles

Select your pre-defined custom role(s). Create a custom role in the Custom Roles tab.

Specific Privileges

Select multiple privileges and what database and collection they are associated with. Leaving collection blank will grant this role for all collections in the database.

Step 5: Menghubungkan Express.js ke MongoDB Atlas

Connect to harits



Connecting with MongoDB Driver

1. Select your driver and version

We recommend installing and using the latest driver version.

Driver	Version
Node.js	6.7 or later

2. Install your driver

Run the following on the command line

```
npm install mongodb
```

[View MongoDB Node.js Driver installation instructions.](#)

3. Add your connection string into your application code

Use this connection string in your application

☐ View full code sample

```
mongodb+srv://harits-2400016087:<db_password>@harits.cgw7hrw.mongodb.net/?
appName=harits
```

Replace `<db_password>` with the password for the `harits-2400016087` database user. Ensure any option params are [URL encoded](#) .

RESOURCES

[Get started with the Node.js Driver](#)

[Node.js Starter Sample App](#)

[Access your Database Users](#)

Troubleshoot Connections

[Go Back](#)

Done

```

(base) macbookpro@MacbookPro:~$ node server.js
(base) macbookpro@MacbookPro:~$ Prak8sbd & npm init -y
Wrote to /Users/macbookpro/Library/Mobile Documents/com-apple-CloudDocs/UA0/SEMESTER 3/PRAKTIKUM/PRAK_SBD/Prak8sbd/package.json:

{
  "name": "Prak8sbd",
  "version": "1.0.0",
  "description": "",
  "main": "server.js",
  "scripts": {
    "test": "echo \"Error: no test specified\"",
    "start": "node server.js"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}

(base) macbookpro@MacbookPro:~$ Prak8sbd & npm install mongodb
added 12 packages, and audited 13 packages in 2s

found 0 vulnerabilities

(base) macbookpro@MacbookPro:~$ Prak8sbd & npm install express
added 65 packages, and audited 78 packages in 2s

22 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

(base) macbookpro@MacbookPro:~$ Prak8sbd & npm install mongoose
npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use it. Check out lru-cache if you want a good and tested way to coalesce async requests by a key value, which is much more comprehensive and powerful.
npm warn deprecated rimraf@3.0.2: Rimraf versions prior to v4 are no longer supported
npm warn deprecated glob@7.2.3: Glob versions prior to v9 are no longer supported
npm warn deprecated we-are-there@yet@2.0.0: This package is no longer supported.
npm warn deprecated gauge@3.0.2: This package is no longer supported.
npm warn deprecated querystring@2.0.0: The querystring API is considered legacy. new code should use the URLSearchParams API instead.
npm warn deprecated google-pem@3.1.4: Package is no longer maintained

added 486 packages, and audited 484 packages in 23s

58 packages are looking for funding
  run `npm fund` for details

11 vulnerabilities (5 moderate, 4 high, 2 critical)

To address issues that do not require attention, run:
  npm audit fix

Some issues may require manual action. To see details, run:
  npm audit --verbose

Some issues require review, and may require choosing a different dependency.

Run `npm audit` for details.

(base) macbookpro@MacbookPro:~$ Prak8sbd &

```

```

(base) macbookpro:MacbookPro-MacBook-Pro Prak8sbd % node server.js
(node:26723) [MONGODB] DeprecationWarning: Mongoose: the 'strictQuery' option will be switched back to 'false' by default in Mongoose 7. Use 'mongoose.set('strictQuery', false);' if you want to prepare for this change. Or use 'mongoose.set('strictQuery', true);' to suppress this warning.
(Use 'node --trace-deprecation ...' to show where the warning was created)
Server is running on http://localhost:3542
Failed to connect to MongoDB MongoServerError: bad auth : authentication failed
    at Connection.onMessage (/Users/macbookpro/Library/Mobile Documents/com~apple~CloudDocs/UAD/SEMESTER 3/PRAKTIKUM/PRAK SBD/Prak8sbd/node_modules/mongoose/node_modules/mongodb/lib/cmap/connection.js:231:30)
    at MessageStream.<anonymous> (/Users/macbookpro/Library/Mobile Documents/com~apple~CloudDocs/UAD/SEMESTER 3/PRAKTIKUM/PRAK SBD/Prak8sbd/node_modules/mongoose/node_modules/mongodb/lib/cmap/message_stream.js:61:60)
    at MessageStream.emit (node:events:508:28)
    at processIncomingData (/Users/macbookpro/Library/Mobile Documents/com~apple~CloudDocs/UAD/SEMESTER 3/PRAKTIKUM/PRAK SBD/Prak8sbd/node_modules/mongoose/node_modules/mongodb/lib/cmap/message_stream.js:125:16)
    at MessageStream.write (/Users/macbookpro/Library/Mobile Documents/com~apple~CloudDocs/UAD/SEMESTER 3/PRAKTIKUM/PRAK SBD/Prak8sbd/node_modules/mongoose/node_modules/mongodb/lib/cmap/message_stream.js:33:9)
    at writeOrBuffer (node:internal/streams/writable:570:12)
    at _write (node:internal/streams/writable:499:10)
    at Writable.write (node:internal/streams/writable:508:10)
    at TLSSocket.ondata (node:internal/streams/readable:1008:24)
    at TLSSocket.emit (node:events:508:28) {
  ok: 0,
  code: 8000,
  codeName: 'AtlasError',
  connectionGeneration: 0,
  Symbol(errorLabels): Set(2) { 'HandshakeError', 'ResetPool' }
}

```

Task for Practice

1. Modifikasi Kode Express.js agar Terhubung ke MongoDB Atlas

```

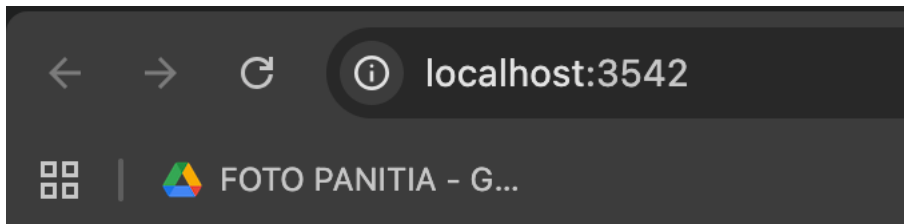
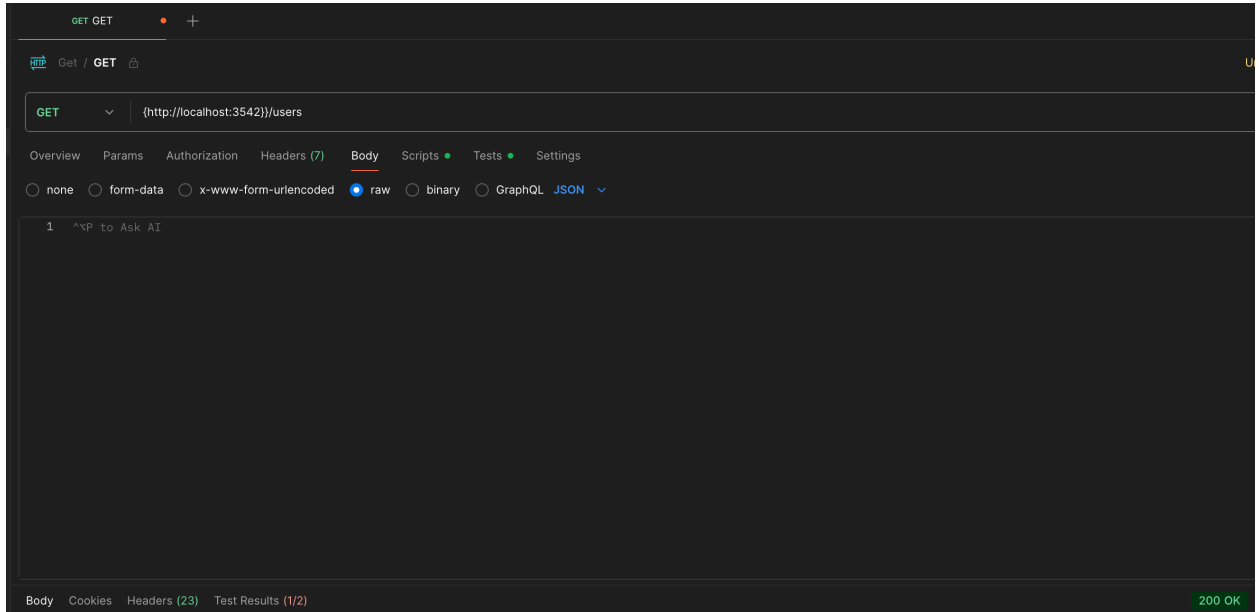
server.js > ...
1  const express = require('express');
2  const mongoose = require('mongoose');
3  require('dotenv').config();
4
5  const app = express();
6  const port = process.env.PORT || 2105;
7
8  // Replace <username> and <password> with your MongoDB Atlas credentials
9  const mongoURI = process.env.MONGO_URI || 'your_mongodb_atlas_uri';
10
11 // Middleware to parse JSON bodies
12 app.use(express.json());
13
14 // Connect to MongoDB
15 mongoose.connect(mongoURI, { useNewUrlParser: true, useUnifiedTopology: true })
16   .then(() => {
17     console.log('Connected to MongoDB');
18   })
19   .catch(err => {
20     console.error('Failed to connect to MongoDB', err);
21   });
22
23 // Sample endpoint to ensure server is running
24 app.get('/', (req, res) => {
25   res.send('Hello, world!');
26 });
27
28 // Endpoint to verify MongoDB connection
29 app.get('/status', async (req, res) => {
30   try {
31     const dbStatus = await mongoose.connection.db.admin().ping();
32     res.json({ status: 'Connected to MongoDB', dbStatus });
33   } catch (err) {
34     res.status(500).json({ status: 'Error', message: err.message });
35   }
36 });
37
38 app.listen(port, () => {
39   console.log(`Server is running on http://localhost:${port}`);
40 });

```

2. Menyetel File Environment (.env)

```
.env
Import to Postman
1 PORT=3542
2 MONGO_URI= mongodb+srv://harits-2400016087:<db_password>@harits.cgw7hrw.mongodb.net/?appName=harits
```

3. Pengujian Koneksi Menggunakan Postman



Hello, world!

C. Kesimpulan

MongoDB Atlas memudahkan proses pembuatan dan pengelolaan database karena berbasis cloud dan tidak memerlukan instalasi manual. Dengan menghubungkan MongoDB Atlas ke aplikasi Node.js menggunakan Express.js dan Mongoose, proses pengolahan data menjadi lebih rapi dan terstruktur, terutama untuk operasi CRUD. Penggunaan file .env juga sangat membantu dalam menjaga keamanan data koneksi database. Selain itu, pengujian menggunakan Postman menunjukkan bahwa koneksi dan

endpoint API dapat berjalan dengan baik. Secara keseluruhan, praktikum ini memberikan pemahaman yang lebih jelas tentang penggunaan database NoSQL serta penerapannya dalam pengembangan aplikasi web.